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Batch: 25

Task 1: Refactoring – Removing Code Duplication

Objective

To eliminate repeated logic by extracting reusable functions.

Task Description

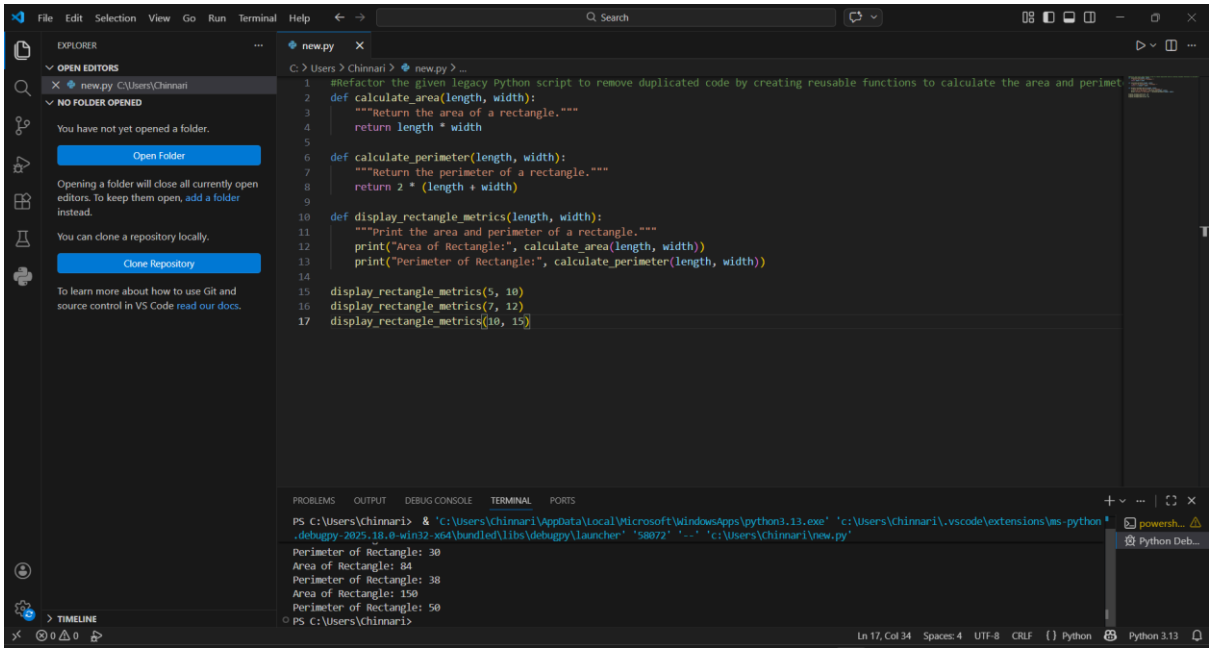
Use AI assistance to refactor a legacy Python script that contains repeated blocks of code calculating the area and perimeter of rectangles.

Starter (Legacy) Code

```
# Legacy script with repeated logic
print("Area of Rectangle:", 5 * 10)
print("Perimeter of Rectangle:", 2 * (5 + 10))
print("Area of Rectangle:", 7 * 12)
print("Perimeter of Rectangle:", 2 * (7 + 12))
print("Area of Rectangle:", 10 * 15)
print("Perimeter of Rectangle:", 2 * (10 + 15))
```

Expected Outcome

- A reusable function to calculate area and perimeter
- No duplicated code blocks
- Proper docstrings for all functions



Task 2: Refactoring – Optimizing Loops and Conditionals

Objective

To improve performance by replacing inefficient nested loops with optimized structures.

Task Description

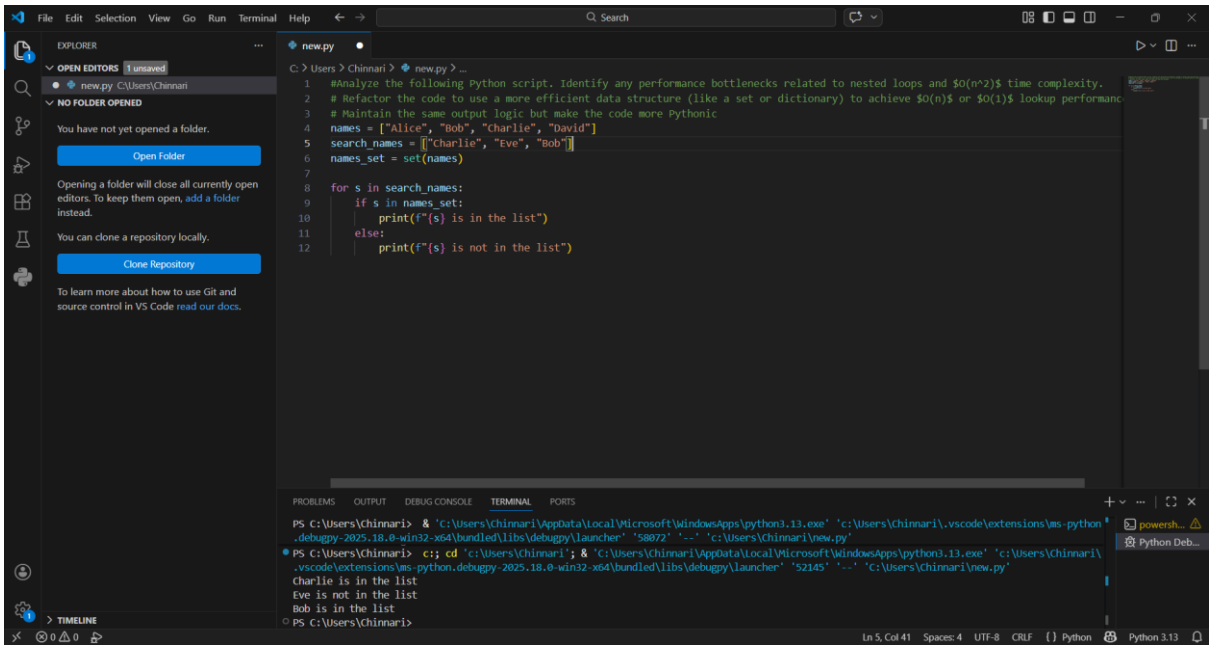
Use AI to analyze and refactor a script that checks the presence of elements using nested loops.

Starter (Legacy) Code

```
names = ["Alice", "Bob", "Charlie", "David"]

search_names = ["Charlie", "Eve", "Bob"]

for s in search_names:

found = False

for n in names:

if s == n:

found = True

if found:

print(f"{s} is in the list")

else:

print(f"{s} is not in the list")
```

Expected Outcome

- Optimized solution using set lookups or comprehensions
- Performance comparison before and after refactoring



Task 3: Refactoring – Extracting Reusable Functions

Objective

To modularize code by extracting calculations into reusable functions.

Task Description

Refactor a legacy script where price and tax calculations are written inline.

Starter (Legacy) Code

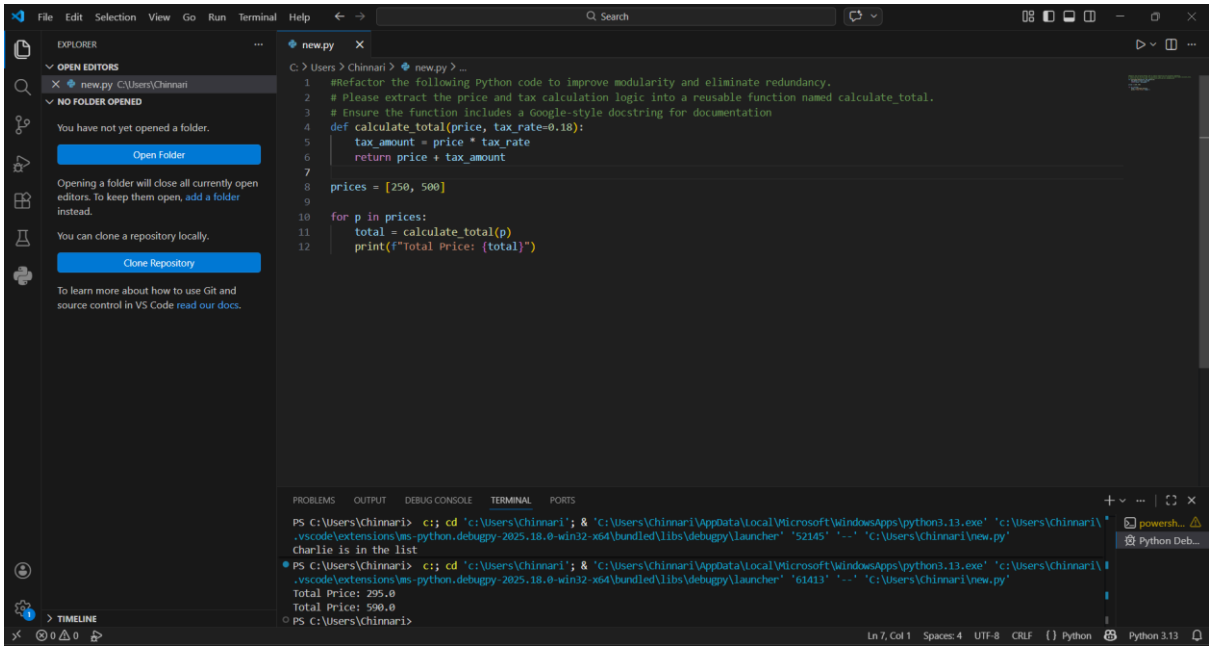
```
price = 250

tax = price * 0.18

total = price + tax

print("Total  Price:", total)

price = 500

tax = price * 0.18

total = price + tax

print("Total  Price:", total)
```

Expected Outcome

• A function calculate_total(price)

• Cleaner and reusable code structure

• Proper documentation



Task 4: Refactoring – Replacing Hardcoded Values with Constants

Objective

To improve maintainability by replacing magic numbers with named constants.

Task Description

Use AI to identify hardcoded values and replace them with constants.

Starter (Legacy) Code

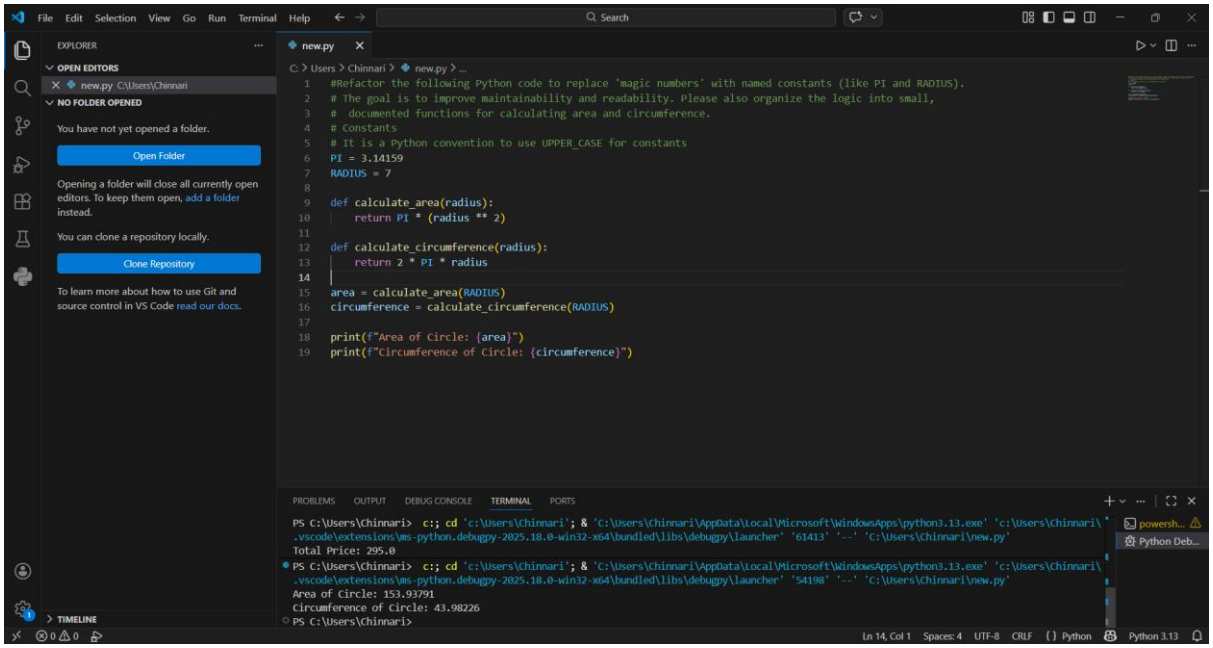
print("Area of Circle:", 3.14159 * (7 ** 2))

print("Circumference of Circle:", 2 * 3.14159 * 7)

Expected Outcome

• Constants such as PI and RADIUS

• Cleaner, maintainable code



Task 5: Refactoring – Improving Variable Naming and Readability

Objective

To enhance readability using descriptive variable names and comments.

Task Description

Refactor a script with unclear variable names.

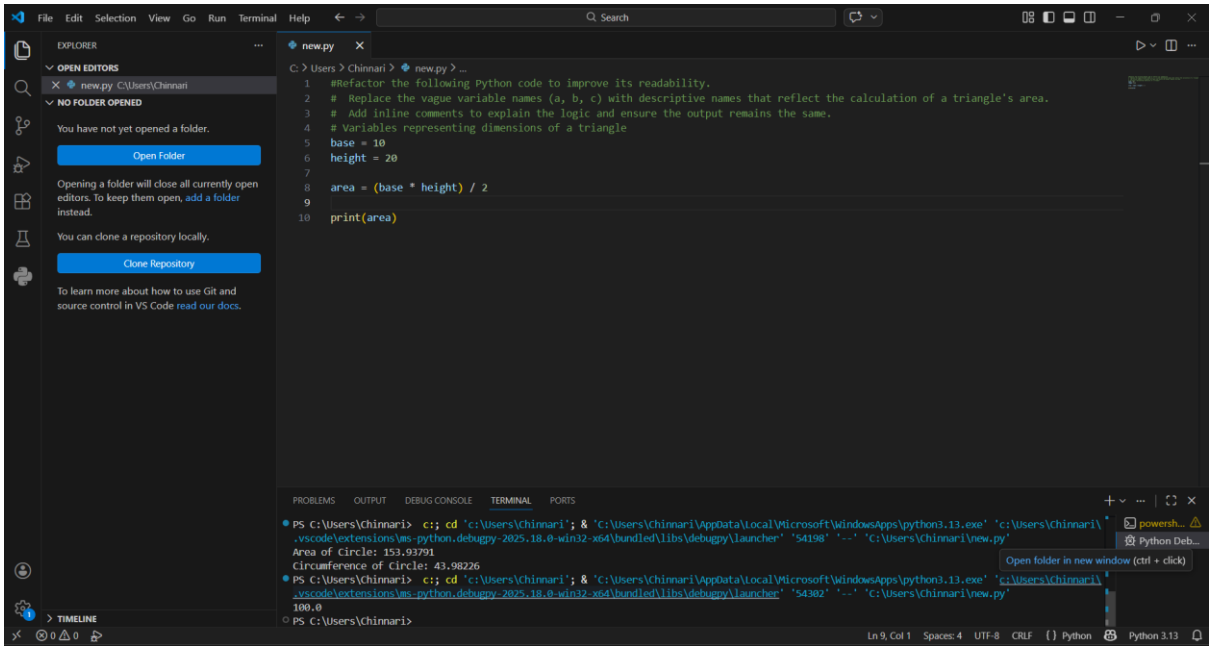
Starter (Legacy) Code

```
a = 10

b = 20

c = a * b / 2

print(c)
```

Expected Outcome

- Descriptive variable names
- Inline comments explaining logic
- Identical output

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.


File Edit Selection View Go Run Terminal Help
Search
EXPLORER
OPEN EDITORS
new.py C:\Users\Chinnari
NO FOLDER OPENED
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
You can clone a repository locally.
Clone Repository
To learn more about how to use Git and source control in VS Code read our docs.
TIMELINE
new.py
C: > Users > Chinnari > new.py > ...
#Refactor the following Python code to improve its readability.
# Replace the vague variable names (a, b, c) with descriptive names that reflect the calculation of a triangle's area.
# Add inline comments to explain the logic and ensure the output remains the same.
# Variables representing dimensions of a triangle
base = 10
height = 20
area = (base * height) / 2
print(area)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Chinnari> c:; cd 'c:\Users\Chinnari'; & 'C:\Users\Chinnari\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\Chinnari\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '54198' '--' 'C:\Users\Chinnari\new.py'
Area of Circle: 153.93791
Circumference of Circle: 43.98226
PS C:\Users\Chinnari> c:; cd 'c:\Users\Chinnari'; & 'C:\Users\Chinnari\AppData\Local\Microsoft\WindowsApps\python3.13.exe' 'c:\Users\Chinnari\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '54302' '--' 'C:\Users\Chinnari\new.py'
100.0
PS C:\Users\Chinnari>
powersh...
Python Deb...
Open folder in new window (ctrl + click)
Ln 9, Col 1 Spaces: 4 UTF-8 CRLF {} Python Python 3.13